

Note on this document:

*This HLD reflects original design intent as of June 2026 — see docs/HLD\_COVERAGE\_MAP.md for how this maps to the actual, as-built system, and docs/CURRENT\_STATE.md for current architecture.*

# HIGH-LEVEL DESIGN DOCUMENT

## Pose Estimation Module

Ergonomic Risk Assessment System

|                          |                 |
|--------------------------|-----------------|
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## 1. Module Objective

The Pose Estimation Module serves as a critical component in the Ergonomic Risk Assessment System. Its primary objective is to convert detected workers from video input into structured body landmarks and biomechanical posture features that enable accurate ergonomic risk assessment.

This module acts as the bridge between visual detection (Member 2) and risk classification (Member 4), transforming raw video data into quantifiable posture metrics using state-of-the-art computer vision techniques.

## Key Performance Indicators

| Metric             | Target Value  |
|--------------------|---------------|
| Body Landmarks     | 33 Points     |
| Feature Dimensions | 15+ per Frame |
| Processing Rate    | ~30 FPS       |
| Latency            | <100ms        |
| Accuracy           | >95%          |

## 2. High-Level Flow Architecture

The Pose Estimation Module operates within a multi-stage pipeline, receiving inputs from the Human Detection Module and providing structured data to the Risk Assessment Module.

| Stage | Input          | Component                | Output         |
|-------|----------------|--------------------------|----------------|
| 1     | Video Frame    | Person Detection (Mohan) | Bounding Box   |
| 2     | Bounding Box   | Pose Estimation Engine   | 33 Landmarks   |
| 3     | 33 Landmarks   | Landmark Validation      | Valid Points   |
| 4     | Valid Points   | Feature Extraction       | Raw Features   |
| 5     | Raw Features   | Angle Calculation        | Angles         |
| 6     | Angles         | Feature Vector Builder   | Feature Vector |
| 7     | Feature Vector | Risk Assessment (Nikhil) | Risk Score     |

### 3. Input Specifications

The Pose Estimation Module receives the following inputs from Member 2 (Mohan - Human Detection Module):

#### 3.1 Bounding Box

- Format: (x, y, width, height)
- Coordinates: Normalized [0.0 - 1.0] or Pixel values
- Confidence Threshold: > 0.7 required
- Purpose: Defines Region of Interest (ROI) for pose estimation

#### 3.2 Tracking ID

- Type: Unique integer identifier
- Persistence: Maintained across frames for the same person
- Lifetime: Valid for entire video sequence
- Purpose: Links pose data to specific worker over time

#### 3.3 Frame Data

| Attribute  | Specification | Notes                |
|------------|---------------|----------------------|
| Format     | RGB Array     | H x W x 3 dimensions |
| Resolution | Min 640x480   | HD recommended       |
| Frame Rate | 30 FPS        | Synchronized         |
| Data Type  | NumPy Array   | uint8 format         |
| Timestamp  | Milliseconds  | Video sync           |

## 4. Processing Pipeline

### Step 1: Pose Estimation (MediaPipe)

The core pose estimation leverages Google's MediaPipe Pose solution to extract 33 anatomical landmarks from the worker's body within the provided bounding box.

#### 4.1.1 MediaPipe Configuration

| Parameter                | Value          |
|--------------------------|----------------|
| Model                    | MediaPipe Pose |
| Landmarks                | 33 Points      |
| Dimensions               | 3D (x, y, z)   |
| Min Detection Confidence | 0.5            |
| Min Tracking Confidence  | 0.5            |

#### 4.1.2 33 Body Landmarks Distribution

| Body Region | Landmarks | Key Points                       |
|-------------|-----------|----------------------------------|
| Face/Head   | 11 points | Nose, eyes, ears, mouth          |
| Arms        | 8 points  | Shoulders, elbows, wrists, hands |
| Torso       | 5 points  | Neck, chest, spine, hips         |
| Legs        | 8 points  | Hips, knees, ankles, feet        |
| Pelvis      | 1 point   | Center of pelvis                 |

### Step 2: Landmark Validation

Before processing, all landmarks undergo rigorous validation to ensure data quality and reliability.

| Check        | Criteria                    | Threshold    | Action                |
|--------------|-----------------------------|--------------|-----------------------|
| Visibility   | Landmark visible in frame   | > 0.5        | Flag as occluded      |
| Confidence   | Model confidence score      | > 0.7        | Interpolate/Reject    |
| Completeness | Essential landmarks present | 25/33 min    | Request re-estimation |
| Coherence    | Anatomical feasibility      | Joint limits | Kalman smoothing      |
| Temporal     | Frame-to-frame stability    | < 15 degrees | Low-pass filter       |

### Step 3: Feature Engineering

Validated landmarks are transformed into biomechanical features critical for ergonomic assessment.

| Feature             | Calculation Method       | Application             |
|---------------------|--------------------------|-------------------------|
| Neck Flexion        | Ear-shoulder-hip angle   | RULA Score Component    |
| Trunk Flexion       | Shoulder-hip-knee angle  | REBA Score Component    |
| Shoulder Elevation  | Vertical shoulder offset | Muscle Load Analysis    |
| Shoulder Symmetry   | L/R shoulder height diff | Postural Balance        |
| Alignment Deviation | Plumb line offset        | Spine Health Assessment |
| Leg Position        | Hip-knee-ankle angles    | Load Distribution       |

### Step 4: Future Features (Research Phase)

Planned enhancements include activity recognition and context-aware analysis for improved risk assessment:

- Movement Pattern Detection: Velocity and acceleration analysis
- Activity Recognition: Lifting, bending, twisting, reaching classification
- Context-Aware Analysis: Task-specific risk rules and environmental factors
- Fatigue Detection: Repetitive motion pattern analysis



## 5. Output Specifications

The Pose Estimation Module delivers the following outputs to Member 4 (Nikhil - Risk Assessment Module):

| Output            | Description                                     | Format  |
|-------------------|-------------------------------------------------|---------|
| Pose Landmarks    | 33 (x, y, z) coordinates with visibility scores | JSON    |
| Body Angles       | Calculated joint angles and deviations          | Array   |
| Feature Vector    | 15+ dimensional normalized feature array        | Vector  |
| Confidence Scores | Per-landmark and overall confidence metrics     | JSON    |
| Tracking ID       | Worker identifier for temporal linking          | Integer |
| Timestamp         | Frame sequence number and time                  | Integer |
| Validation Flags  | Data quality indicators                         | Boolean |

## 6. Dependencies

### 6.1 Inputs From (Upstream)

- **Member 2 (Mohan): Human Detection Module**
  - Bounding Boxes (coordinates and dimensions)
  - Tracking IDs (persistent worker identifiers)
  - Frame Data (video frames and metadata)

### 6.2 Outputs To (Downstream)

- **Member 4 (Nikhil): Risk Assessment Module**
  - Feature Vectors (structured pose data)
  - Biomechanical Angles (ergonomic metrics)
  - Confidence Scores (data quality metrics)

### 6.3 Interface Specifications

| Interface       | Specification                       |
|-----------------|-------------------------------------|
| Input Protocol  | Shared memory / Redis message queue |
| Output Protocol | JSON via RabbitMQ message broker    |
| Data Format     | JSON with base64-encoded arrays     |
| Frequency       | Real-time (30 FPS)                  |

## 7. Risk Analysis and Mitigation

The following risks have been identified with corresponding mitigation strategies:

| <b>Risk</b>         | <b>Impact</b> | <b>Likelihood</b> | <b>Mitigation Strategy</b>                  |
|---------------------|---------------|-------------------|---------------------------------------------|
| Occlusion           | High          | Medium            | Multi-camera fusion, temporal interpolation |
| Poor Lighting       | High          | Medium            | HDR preprocessing, IR camera backup         |
| Multiple Workers    | Medium        | High              | Tracking ID isolation, depth ordering       |
| Missing Keypoints   | Medium        | Medium            | Kalman filtering, anatomical constraints    |
| Industrial PPE      | Medium        | High              | PPE-aware training data                     |
| Compute Limitations | Low           | Low               | GPU acceleration, model quantization        |

## 8. Future Enhancements Roadmap

The Pose Estimation Module is designed for extensibility. The following roadmap outlines planned enhancements:

### Current State

**Pose Estimation -----> Risk Assessment**

### Future State (Research Phase)

**Pose Estimation -----> Activity Recognition -----> Context Analysis -----> Human Validation -----> Risk Assessment**

### Enhancement Details

- Activity Recognition: LSTM/Transformer-based classifier to identify specific worker activities (lifting, bending, reaching, twisting)
- Movement Pattern Detection: Velocity profiles, acceleration analysis, and repetition counting for repetitive task identification
- Context-Aware Analysis: Integration of task context, environmental factors, and zone-based risk rules
- Human-in-the-Loop Validation: Expert review interface with semi-supervised learning for continuous model improvement
- Mobile Integration: Real-time supervisor alerts via mobile application

## 9. Technical Stack

The Pose Estimation Module utilizes the following technology stack:

| Category         | Technology       | Purpose                    |
|------------------|------------------|----------------------------|
| Core Engine      | MediaPipe Pose   | Pose landmark detection    |
| Language         | Python 3.9+      | Primary development        |
| Image Processing | OpenCV, NumPy    | Frame manipulation         |
| ML Framework     | TensorFlow Lite  | Optimized inference        |
| Message Queue    | Redis / RabbitMQ | Inter-module communication |
| Database         | InfluxDB         | Time-series storage        |
| Deployment       | Docker           | Containerization           |
| Hardware         | NVIDIA GPU       | Acceleration               |

## 10. Performance Metrics

The module is expected to achieve the following performance benchmarks:

| Metric               | Target            |
|----------------------|-------------------|
| Processing Rate      | ~30 FPS           |
| End-to-End Latency   | <100 milliseconds |
| Landmark Accuracy    | >95%              |
| Validation Pass Rate | >85%              |
| Memory per Worker    | ~50 MB            |
| Concurrent Workers   | 10+ (scalable)    |
| CPU Utilization      | <70% peak         |
| GPU Utilization      | <80% peak         |